

SEMESTER VIII

Programme	B. Sc. STATISTICS				
Course Code	STA 8 CJ 406 (P)				
Course Title	APPLIED STOCHASTIC PROCESSES AND TIME SERIES ANALYSIS				
Type of Course	Major				
Semester	VII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practicum per week	Total Hours
	4	3	-	2	75
Pre-requisites	Basic knowledge of Markov chain & general aspects of time series				
Course Summary	Understand queue, renewal process and Brownian process. Thorough knowledge about auto-correlation and autoregressive moving average.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
1.	Discuss the basic characteristic of a queuing system and acquire skills in analyzing queuing models	U	F	Seminar Presentation / Group Tutorial Work
2.	Analyze a network of queues	An	P	Instructor-created exams
3.	Describe the concept of renewal process and Brownian motion	U	F	Instructor-created exams
4.	Describe the basics of time series data, its auto-covariance, auto-correlation and autoregressive moving average	U	F	Instructor-created exams / Home Assignments
5.	Learn to validate a model using residual analysis	An	P	Home Assignment
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Continuous time Markov chains and Queueing theory.		14	20
	1	Continuous Time Markov Chains		
	2	Pure birth process, Yule furry process, Pure death process, Birth and Death Processes, The transition probability function, Limiting probabilities		
	3	Introduction to queueing theory, Steady state probabilities.		
	4	Exponential Models: A single server Exponential queueing system, A single server Exponential queueing system having finite capacity, Birth and Death queueing models		
	5	Network of queues: Open systems, Closed systems		
	6	Non Markovian queueing models: M/G/1 and G/M/1		
II	Renewal process and Brownian motion		9	15
	7	Renewal processes, renewal function and renewal density, renewal equation, stopping time		
	8	Wald's equation, limit theorems and their applications.		
	9	Brownian motion-Definition, limiting form of random walk, examples.		
	10	White noise, Gaussian process		
	11	Strictly stationary and weakly stationary processes (Definition and examples)		
	12	Branching process (Concept only)		
III	Time series and stationary process		12	20
	13	Time series as a discrete parameter stochastic process		
	14	Auto – Covariance, Auto- Correlation		
	15	Autoregressive, Moving Average, Autoregressive Moving Average and Autoregressive Integrated Moving Average Models		
	16	Choice of AR / MA periods		
	17	Introduction to non-linear time Series: ARCH and GARCH models		
IV	Estimation of ARMA models, ,.		10	15
	18	Yule – Walker estimation for AR Processes		
	19	Maximum likelihood and least squares estimation for ARMA Processes		
	20	Discussion (without proof) of estimation of mean, Auto-covariance and auto-correlation function under large samples theory		
	21	Residual analysis and diagnostic checking		
	22	Forecasting using ARIMA models		
V	Open Ended		30	
		Problems, examples and analysis of dataset using software		

Text Books

1. **Ross, S.M. (2007).** Introduction to Probability Models. IXth Edition, Academic Press.
2. **Medhi, J. (1996).** Stochastic Processes. Second Editions. New Age International
3. **Box G.E.P and Jenkins G.M. (1994).** Time Series Analysis, Forecasting and Control. Holden-Day
4. **Brockwell P.J. and Davis R.A. (2006).** Time Series: Theory and Methods, Springer – Verlag.
5. **Abraham B and Ledolter J.C. (1983).** Statistical Methods for Forecasting, Wiley
6. **Robert H Shumway and Davis S Stoffer (2016).** Time series analysis and its applications with R examples. Springer.

References

1. **Karlin, S. and Taylor, H.M. (1975).** A First Course in Stochastic Processes, Second Edition, Academic Press.
2. **Cinlar, E. (1975).** Introduction to Stochastic Processes. Prentice Hall. New Jersey.
3. **Basu, A.K. (2003).** Introduction to Stochastic Processes. Narosa, New-Delhi
4. **Anderson T.W (1971).** The Statistical Analysis of Time Series, Wiley.
5. **Fuller W.A. (1978).** Introduction to Statistical Time Series, John Wiley
6. **William W. S. Wei (2006).** Time Series Analysis: Univariate and Multivariate Methods. Pearson. Addison Wesley.